



Value Engineering Cases Electronics

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VALUE ENGINEERING CASE EXAMPLES

VAVE - Automotive Power Steering

Business Challenges

- Customer losing competitiveness and business due to high cost
- Multiple PC boards along with flat cables for interconnection
- Complex Engineering arrangement and higher manufacturing cost
- Reliability concerns
- Sub-optimal design elements getting into other power steering variants



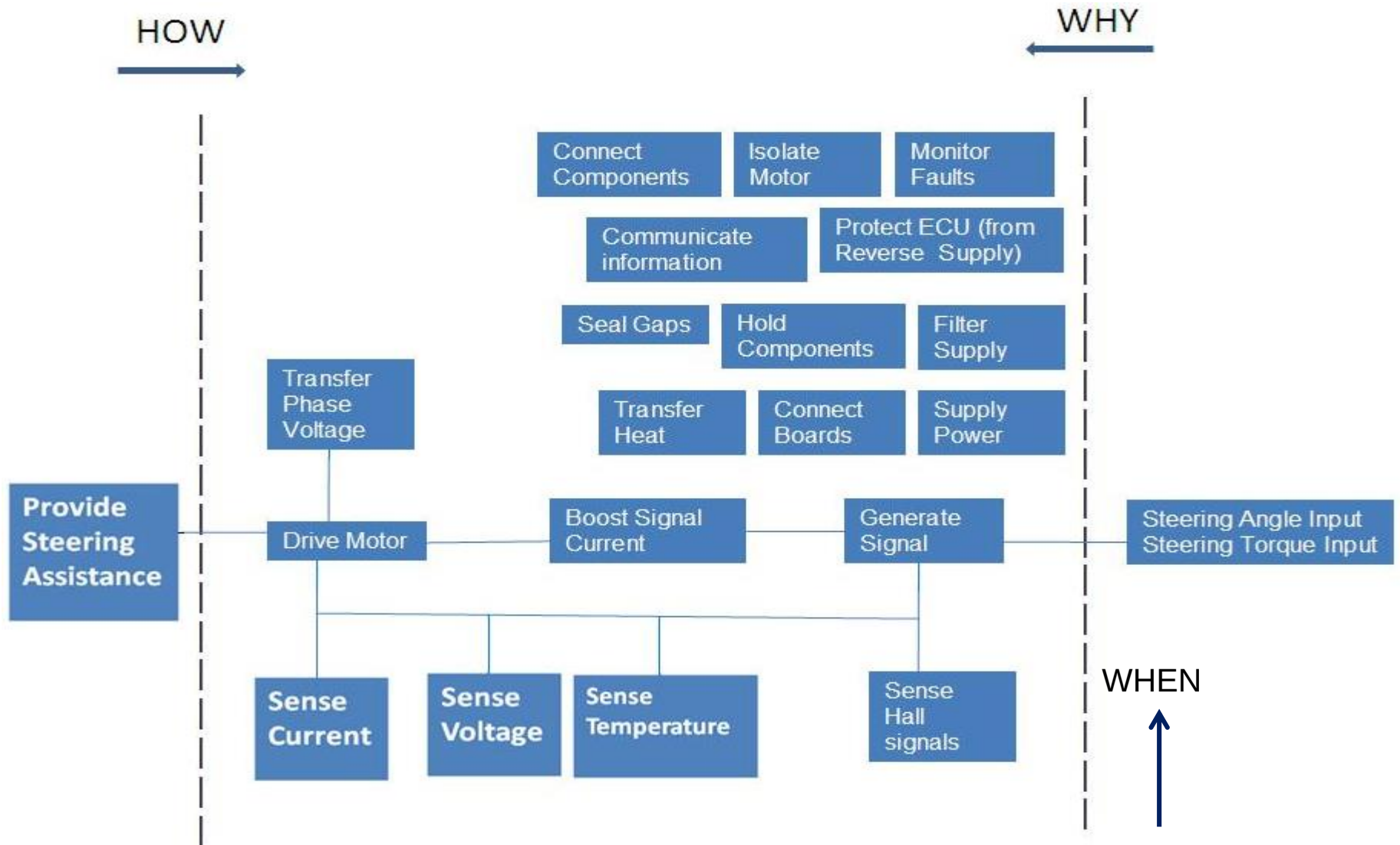
Information Phase (Data Collection / Study)

- Hardware architecture
- Schematic, Costed Bill of Materials
- Product Packaging
- Engineering drawings
- Validation requirements
- Performance specifications
- Manufacturing processes used
- Supply chain information
- Any challenge / issue faced



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Functional Phase - FAST Diagram



Function – Costing

#	Function Name	Cost (\$)
1	Drive Motor	13.00
2	Protect ECU	1.27
3	Sense Current	1.50
4	Sense Voltage	1.50
5	Monitor Faults	1.89
6	Supply Power	4.62
7	Boost Signal Current	2.62
8	Sense Temperature	0.81
9	Sense Hall Signals	0.81
10	Hold Components	1.48
11	Connect Boards	3.41
12	Seal Gaps	3.84
13	Connect Components	1.45
14	Transfer phase voltage	1.63
15	Isolate Motor	1.27
16	Filter supply	1.47
17	Generate Signal	0.81
18	Transfer Heat	3.03
19	Communicate Information	0.32
20	Others	3.75

Function-Cost-Worth Analysis

#	Function Name	Cost (\$)	Worth (\$)	Comments on Worth
1	Drive Motor	13.00	0.40	Considered a simple DC motor driver for calculating worth
2	Protect ECU	1.27	0.04	Diode
3	Sense Current	1.50	0.10	Micro
4	Sense Voltage	1.50	0.10	Micro
5	Monitor Faults	1.89	0.15	Voltage regulator with watch dog functionality
6	Supply Power	4.62	0.40	2 regulators
7	Boost Signal Current	2.62	0.30	Transistor based driver pack
8	Sense Temperature	0.81	0.10	Micro
9	Sense Hall Signals	0.81	0.10	Micro
10	Hold Components	1.48	0.20	Acrylic sheet
11	Connect Boards	3.41	0.50	Connector (Actual: 16way interconnect assembly , 12way interconnect assembly and connector ,12way connector)
12	Seal Gaps	3.84	1.00	Sealant (Existing: Gasket)
13	Connect Components	1.45	0.40	Wires
14	Transfer phase voltage	1.63	0.50	Small bus bar
15	Isolate Motor	1.27	0.40	Single pole relay
16	Filter supply	1.47	0.60	LC filter
17	Generate Signal	0.81	0.60	Micro
18	Transfer Heat	3.03	2.40	Thermal Pad and thermal paste (Existing: Base Plate, Thermal Paste 3ml, Capacitor heat sink)
19	Communicate Information	0.32	0.32	
20	Others	3.75		

Function-Cost-Worth Analysis

#	Function Name	Cost (\$)	Worth (\$)	Value Index (Cost/Worth)	Comments on Worth
1	Drive Motor	13.00	0.40	32.50	Considered a simple DC motor driver for calculating worth
2	Protect ECU	1.27	0.04	31.75	Diode
3	Sense Current	1.50	0.10	15.00	Micro
4	Sense Voltage	1.50	0.10	15.00	Micro
5	Monitor Faults	1.89	0.15	12.61	Voltage regulator with watch dog functionality
6	Supply Power	4.62	0.40	11.55	2 regulators
7	Boost Signal Current	2.62	0.30	8.73	Transistor based driver pack
8	Sense Temperature	0.81	0.10	8.14	Micro
9	Sense Hall Signals	0.81	0.10	8.14	Micro
10	Hold Components	1.48	0.20	7.40	Acrylic sheet
11	Connect Boards	3.41	0.50	6.82	Connectors (Existing: 16way interconnect assembly , 12way interconnect assembly and connector,12way connector)
12	Seal Gaps	3.84	1.00	3.84	Sealant (Existing: Gasket)
13	Connect Components	1.45	0.40	3.63	Wires
14	Transfer phase voltage	1.63	0.50	3.26	Small bus bar
15	Isolate Motor	1.27	0.40	3.18	Single pole relay
16	Filter supply	1.47	0.60	2.46	LC filter
17	Generate Signal	0.81	0.60	1.36	Micro
18	Transfer Heat	3.03	2.40	1.26	Thermal Pad and thermal paste (Existing: Base Plate, Thermal Paste 3ml, Capacitor heat sink)
19	Communicate Information	0.32	0.32	1.00	
20	Others	3.75			

Creative Phase

- Use of creative techniques to identify other ways to perform the system function(s).
- Focus on the quantity of ideas



Creative Phase - Generation of Alternate Ideas through Brainstorming

#	Drive Motor	Protect ECU	Monitor Faults	Supply Power
1	Discrete MOSFET	Fuse and diode	Watch dog in power supply	Use linear regulators
2	MOSFETs in wafer form	Single way connecter	Sensors and actuator	SMPS
3	Transistors	Switch	Inbuilt watch dog in ASIC/ remove monitor micro	Functional generator
4	CMOS disc drive	Transistor as a switch	Safety ASIC	Use 5v battery
5	Relays	Solid state relay	Microprocessor based steering angle sensor	Multi output regulator
6	Direct drive	Surge protector	No monitoring	DC to DC power supply
7	Optical drive	Connector to the battery	Four angle sensor	Varta batteries
8	DC to DC converter	Relay	Use watchdog timer	Amplifier and attenuators
9	AC to DC converter	MOSFETs	Relay	Heat to voltage converter
10	Sterling engine	SCR	555 timer	Solar energy
11	High voltage motor	Zener diode	Hard logic for disabling mechanism	Sterling engine
12	Switched reluctance	Reed switch	Small microcontroller	Super capacitor
13	Change motor	Inductor	Fail safe mechanism	Back EMF
14	No motor	DC to DC Converter		Wind energy

Evaluation Phase - Analysis and Short listing of Ideas

- Identify the Feasible ideas
- Costing, Choosing by Advantages (CBA), Value Metrics
 - Analyzing the ideas on various parameters (cost, quality, performance, reliability, customer acceptance)
- Short listing of Best Ideas



Evaluation Phase- Short listing of Feasible Ideas

#	Drive Motor	Protect ECU	Monitor Faults	Supply Power	Boost Signal Current
1	Discrete MOSFETs	Fuse And Diode	Watch Dog In Power Supply	Use Linear Regulators	Buffer IC's
2	MOSFETs In Wafer-form	Single Way Connector	Sensors And Actuator	SMPS	Amplifier
3	Transistors	Switch	Inbuilt Watch Dog In ASIC/ Remove Monitor Micro	Functional Generator	Transistor
4	CMOS Disc Drive	Transistor As A Switch	Safety ASIC	Use 5V Battery	ASIC
5	Relays	Solid State Relay	Microprocessor Based Steering Angle Sensor	Multi Output Regulator	Moving Low Power Electronic Components To ASIC

Feasible Ideas

Rejected Ideas

Evaluation Phase - Value analysis of Key Ideas

Criteria	Cost	Quality	Reliability	Performance	Customer Acceptance	Criteria Value
		B	C	D	E	
Cost	A	B-3, A-2	C-3, A-2	D-2, A-2	E-3, A-1	7
Quality		B	C-2, B-2	B-3, D-2	E-3, B-1	9
Reliability			C	D-2, C-2	E-2, C-1	8
Performance				D	E-2, D-1	7
Acceptance of customer					E	10

A	Cost
B	Quality
C	Reliability
D	Performance
E	Acceptance of customer

Criteria Table

	Functions	Drive Motor											Protect ECU											
	Criteria Values	7		9		8		7		10				7		9		8		7		10		
#	Ideas	Cost		Quality		Reliability		Performance		Customer Acceptance		Total	Ideas	Cost		Quality		Reliability		Performance		Customer Acceptance		Total
1	Discrete MOSFETs	5	35	5	45	4	32	5	35	5	50	197	Fuse and diode	5	35	5	45	5	40	5	35	4	40	195
2	Motor with built in MOSFETs	4	28	4	36	3	24	5	35	2	20	143	Link Relay	2	14	5	45	4	32	4	28	5	50	169
3	Danfoss Power module	2	14	5	45	5	40	5	35	2	20	154	Solid state relay	4	28	4	36	4	32	4	28	5	50	174
4													Opto isolated MOSFETs	2	14	5	45	2	16	4	28	3	30	133
5													Polarized connectors	3	21	5	45	1	8	1	7	2	20	101

#	Function Name	Cost (\$)	Reworked cost (\$)	Savings (\$)	Comments on Worth
1	Drive Motor	13.00	6.00	7.00	Using Discrete MOSFET
2	Protect ECU	1.27	0.70	0.57	Body diode of MOSFET can be used for reverse battery protection.
3	Monitor Faults	1.89	0.95	0.94	PIC Controller - PIC16F690
4	Supply Power	4.62	3.12	1.62	Regulator without SPI
5	Connect Boards	3.41	0.10	3.31	(Existing: 16way interconnect assembly, 12way interconnect assembly and connector, 12way connector) 10 pin Berg stick used.
6	Seal Gaps	3.84	1.92	1.92	Cardboard gasket (50% cost considered) + Sealant
7	Transfer phase voltage	1.63	0.50	1.13	small bus bar
8	Transfer Heat	3.03	2.10	0.93	Thermal Pad (1.2) and thermal paste (0.9) (Existing: Base Plate 1.1, Thermal Paste 3ml 1.81), Capacitor heat sink (0.1239)
9	Integrate components	6.51	3.69	2.82	Include all the low power discrete components in to ASIC (includes Boost signal current)

Development, Presentation & Implementation Phase.

- **Development Phase**

- The review of ideas by experts, and shortlist the ideas that can be implemented.

- **Presentation Phase**

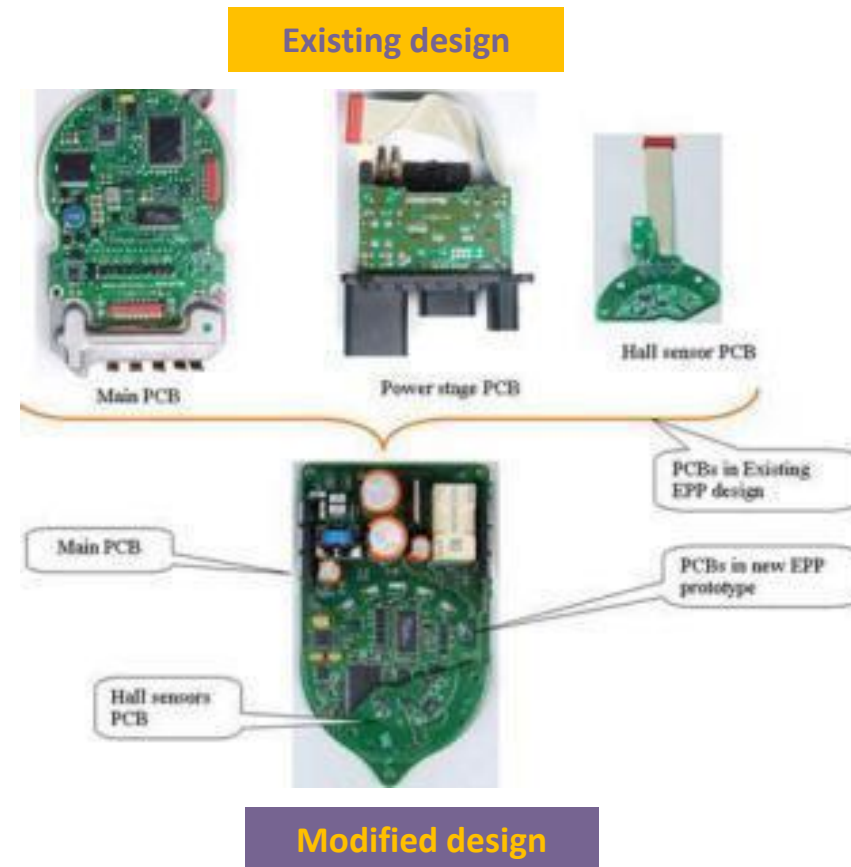
- Presentation of the business case to the higher management to get the approval for the implementation of the ideas.

- **Implementation Phase**

- A detailed work process is planned to implement the ideas.
- Approved ideas implemented during this phase
- Prototypes tested and validated in the lab

Benefits to the customer

- Proposed VE ideas/savings. Customer agreed for implementation of savings per ECU to the tune of 10% in Electronics and 8.8% weight reduction mechanically
- Cables eliminated thereby improving the reliability
- Improved thermal efficiency
- Reduced the number of boards
- Simplified design and use of more standard parts
- Commonisation of EPP for RH drive and LH drive vehicles thereby reducing the inventory requirements
- Decrease in weight equal to 8.8%
- Value Added concepts from FMEA & Reliability Prediction
- Manufacturing Process reduction thereby improving Time to market
- Ideas taken for use in other power steering variants



Thank you

Safe Harbor

This document contains forward-looking statements within the meaning of Section 27A of the Securities Act of 1933, as amended, and Section 21E of the Securities Exchange Act of 1934, as amended. The forward-looking statements contained herein are subject to certain risks and uncertainties that could cause actual results to differ materially from those reflected in the forward-looking statements. Tech Mahindra and Mahindra Satyam undertakes no duty to update any forward-looking statements.